

JOSIE'S 7-TRIP FIELD WORKBOOK

First Flake

A Field Workbook for Beginner Gold Prospectors



***From your first creek to your first flake of gold,
in seven trips.***

JOSIE McCALL, 28, FIELD GEOLOGIST

with notes from Grandpa Ezra, 100

Read the river. It already knows where the gold is.

Contents

GETTING READY

A Letter from Josie	3
Why This Works	4
Before You Start	5
What to Expect	6
The Empty Pan Rule	7
Make It Fit Your Life	8
Safety & Permission Basics	9

THE SEVEN TRIPS

Trip 1: The Tub Rehearsal	10
Trip 2: Read the River	12
Trip 3: Down to Bedrock	14
A Letter from Grandpa Ezra	16
Trip 4: Classify & Concentrate	17
Trip 5: Black Sand School	19
Trip 6: The Sample Line	21
Trip 7: First Flake Day	23
Look How Far You Came	25

YOUR FIELD KIT

The Creek Card	27
Gear Checklist	28
Blank Sample Log	29
Bonus 1: Why Is My Pan Empty?	30
Bonus 2: Ezra's 10 Old-Timer Rules	31
Bonus 3: Prospector Talk	32
Bonus 4: The One-Day Plan	33
The Honest Small Print	34

A Letter from Josie

Hi, friend.

If you're holding this, you've probably already bought a gold pan. Maybe it's still in the plastic. Maybe you've tried it once, washed a whole afternoon of gravel back into the creek, and gone home with sore knees and nothing to show for it. Both are fine. Everyone starts somewhere, and nobody starts good. I certainly didn't.

Here's what I wish someone had told me sooner: finding gold isn't luck. It's reading. Rivers sort their gravel the same way every single time, and once you know the pattern, you stop digging at random and start digging where the river already put it.

That's what this workbook does. Seven trips. Each one teaches one skill, gives you one rule to remember, and has a page where you write down what actually happened. By Trip 7 you won't be hoping anymore. You'll know where to look, how to get it out, and how to prove it's real.

A couple of honest things before we go. You're not going to get rich. Most of what you'll find is flakes smaller than a sesame seed. But I promise you, the first time one of those flakes sits in the bottom of your pan, glowing in the shade, you will yell. Everyone yells. I've watched grown men in hip waders do it.

My great-grandpa Ezra has been panning since 1939. He'll be popping up in the margins with his two cents. He insisted. (He's 100. You don't argue with a man who's 100.)

Grab your pan. Let's go read a river.

Josie

Read the river. It already knows where the gold is.

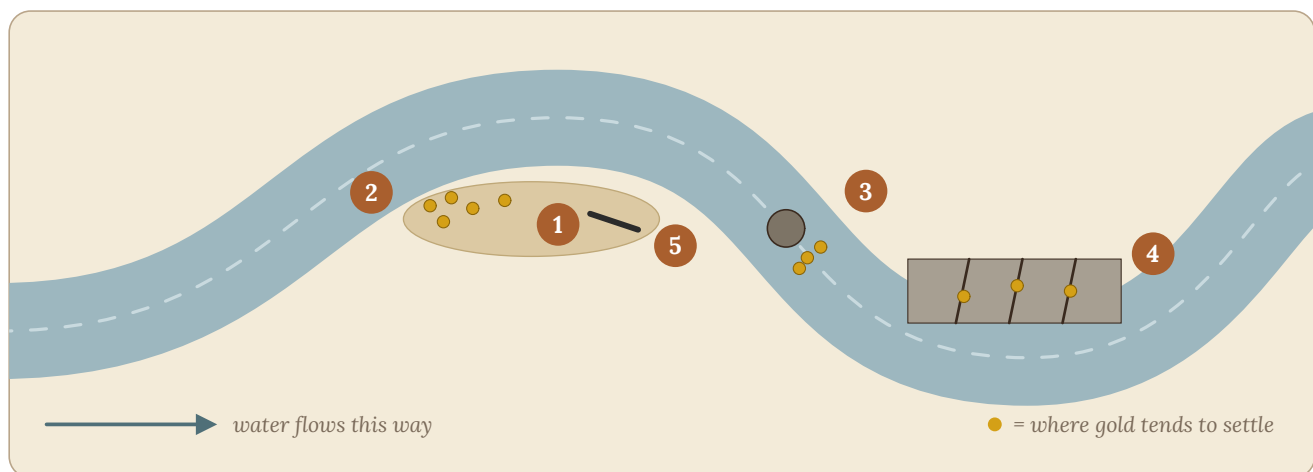
Why This Works

Rivers are sorting machines. Learn how they sort.

Gold is heavy. Really heavy. A flake of gold weighs about seven times as much as a grain of ordinary sand the same size. (Gold's density is about 19.3 grams per cubic centimeter. Quartz sand is about 2.65.) That one fact is the whole science of this book.

When a creek floods, it picks up sand, gravel and gold and carries them downstream. The moment the water slows down, it drops what it can't carry anymore, and it drops the heaviest things first. So gold ends up in the same kinds of places, creek after creek:

1. **The inside of bends**, where the water slows on the curve
2. **The head of gravel bars**, the upstream end where the bar begins
3. **The quiet water behind big boulders**, on the downstream side
4. **Cracks and low spots in bedrock**, where gold sinks and gets trapped
5. **Streaks of black sand**, the heavy minerals that travel with gold



Old-timers learned this by trial and error over whole lifetimes. Geologists call it placer deposition. You're going to learn it in seven trips, by looking first and digging second.

What this workbook is not

- **Not a promise of riches.** Recreational panners usually find small flakes. Their real value is in the finding.
- **Not legal advice.** Where you can pan depends on your state and the type of land. The Safety & Permission page shows you how to check.
- **Not a shortcut.** The skill lives in your hands, and hands learn by doing. That's why every trip ends with a page you fill in.

YOUR STARTING LINE

Before You Start

Score yourself honestly today. You'll score yourself again after Trip 7.

What I'm rating (1 to 10)	Today	After Trip 7
How confident I am about where to dig in a creek		
How good my panning is, without losing what I find		
How sure I am about where it's legal to pan		
How well I can tell real gold from fake		

A note to my future self. Write it now, read it after Trip 7:

Date I started: _____

What to Expect

The road to your first flake has a dip in it. It's on the map.

When	What it feels like	What's really happening
Trips 1 and 2	The fun part	New gear, new skills, and a lot of "oh, so that's how it works." Enjoy it.
Trip 3	The dip	Bedrock cracks are hard work, and plenty of them will be empty. This is exactly where most beginners quit. You won't, because you knew it was coming.
Trips 4 and 5	The click	Once you start saving concentrates and seeing black sand, the pan suddenly makes sense.
Trip 6	The map	Your sample pans start pointing somewhere. The numbers go up in one direction.
Trip 7	The flake	You go back to the best spot and stay there. This is the day you yell.

How long does it take? Most people do one trip per weekend, so about two months. Faster is fine. Slower is fine. That gold has been sitting in the creek for thousands of years. It'll wait for you.

My trip calendar

Trip	Planned date	Done date	Where I went
Trip 1			
Trip 2			
Trip 3			
Trip 4			
Trip 5			
Trip 6			
Trip 7			

The Empty Pan Rule

You will have empty pans. Probably whole empty days. Here's the rule for those days:

No restarts. No catching up. No doubling up. Just do the next trip.

Missed a weekend? Do the next trip. Rained out for a month? Do the next trip. Went to the creek and found nothing? Write it in the log, then do the next trip.

An empty pan isn't a failure. It's information. It tells you where the gold isn't, and in prospecting that's half of knowing where it is. Every real prospector has a notebook full of zeros. The difference is, they kept the notebook.

The log is a map, not a report card.

Ezra says: "I went forty days without a single flake in 1949. Day forty-one paid for every one of them."

My zero-day tally

Every trip that comes home empty gets a mark here. Old-timers were proud of theirs. So be proud of yours.

Make It Fit Your Life

The seven trips are the default. Here's how to bend them without breaking them.

No creek nearby?

Buy a bag of paydirt (real gold-bearing gravel sold for practice) and do Trips 1, 4 and 5 at home in a tub. Many states also have public panning areas and pay-to-pan spots where beginners are welcome. Save Trips 2, 3, 6 and 7 for when you can get to real water.

Bringing kids?

Give every kid a job: sieve captain, BB counter, flake spotter. Keep them in water below the knee and away from anything fast. Cold water tires small bodies quickly, so plan short sessions and pack dry socks.

Knees, back or balance not what they were?

You don't have to kneel in the creek. Sit on a bucket with a lid, work from the bank, and let a classifier do the heavy lifting. A long-handled scoop saves your back. Panning in a tub on a picnic table is still panning.

Only have one free day?

Start with the One-Day Plan at the back of this book. Then come back and do the trips properly once you're hooked. You will be.

Confused about the rules where you live?

Call your local Bureau of Land Management (BLM) field office or Forest Service ranger district. They answer these questions every day, and asking them is exactly what experienced prospectors do. It's the smart move, not the beginner move.

Safety & Permission Basics

Stay safe

- **Cold water is the real danger.** Mountain creeks can numb your hands in minutes, even in July. Take breaks, wear gloves, and bring dry clothes.
- **Check the weather upstream,** not just where you're standing. Rain in the mountains can raise a creek fast. If the water rises or turns muddy, get out.
- **Never enter old mine tunnels or shafts.** They collapse and can hold bad air. Stay out, stay alive.
- **Don't dig into creek banks** or under overhangs. They can come down on you.
- **Tell someone** where you're going and when you'll be back. Better yet, bring a friend.
- **Never use mercury.** Old-timers did. It poisons you and the river, and you don't need it.

Stay legal (a general guide, not legal advice)

- **Private land:** you need the owner's permission. Always.
- **National parks:** prospecting and collecting are generally not allowed.
- **BLM land and national forests:** casual panning is often allowed, but not on ground someone has already claimed. An active mining claim gives the claim holder the mineral rights, even on public land.
- **Checking for claims:** the BLM's online Mineral & Land Records System (MLRS) lets you look up mining claims, and your local BLM office can help you read it.
- **State rules vary.** Some states limit tools, seasons, or digging in fish habitat. Check your state's natural resources or fish and wildlife agency before you go.
- **Leave it better than you found it.** Fill your holes, put rocks back, pack out your trash. That's how panning stays allowed.

The Tub Rehearsal

The Tub Rule: Lose it at home, so you never lose it in the river.

Where: your backyard, garage or bathtub · **Time:** about 45 minutes

Why this trip comes first

Every beginner I've ever taught makes the same mistake. They drive two hours to a creek, pan like they're stirring soup, and wash their gold straight back into the water without ever seeing it. You can't learn a new motion and read a river at the same time. So today, the river comes to you.

Here's the trick: gold is about seven times heavier than the sand and gravel around it. Anything heavy enough will behave the same way in your pan. So we practice with something you can count.

What to bring

- Your gold pan (the riffles are the ridges along one side)
- A plastic tub or bin, wider than your pan, filled with water
- Two big scoops of gravel and sand from your yard or a garden store
- 10 steel BBs or small fishing sinkers. These are your "gold." Wash your hands afterward if you use lead.

At the creek: step by step

1. Mix your 10 BBs into the gravel, then fill your pan about three-quarters full.
2. Sink the whole pan under the water. Break up any clumps with your fingers until everything is loose and muddy.
3. Shake the pan hard, side to side, still under the water, for about 20 seconds. This is the part everyone rushes. Don't. The shaking is what sends heavy things to the bottom.
4. Pick out the big rocks by hand. Glance at each one, then toss it.
5. Tilt the pan forward, riffles on the far side, so the edge dips just under the surface. Swish gently forward and back so the water carries the top layer out over the edge.
6. Every few swishes, level the pan and shake again. Shake, tilt, wash. Shake, tilt, wash.
7. When you're down to a few spoonfuls, swirl slowly and count your BBs.

Trip 1 Field Log

Your goal: Find all 10 BBs, three rounds in a row.

My practice log

Round	BBs found (of 10)	What went wrong?
1		
2		
3		
4		
5		
6		

Josie's field note

The first time I did this, I was five and Grandpa made me use his old coffee can lid as a pan. I lost six out of ten BBs and cried. He just laughed and handed me the bucket again. You'll lose some too. That's the whole point of practicing at home.

Ezra says: "A man who can't find a BB in his own backyard ain't finding gold in the Yukon."

What I noticed: _____

☐ **Trip 1 complete. I found all 10 BBs, three rounds in a row.**

Read the River

The Inside Bend Rule: Gold stops where the water gets lazy.

Where: a creek where you're allowed to pan · **Time:** about 2 hours

Why we look before we dig

Today you'll spend more time looking than digging. That will feel wrong. Do it anyway.

The river has already done the hard work of sorting its gravel. Your job is to read where it put the heavy stuff, and you can do most of that reading from the bank before your boots even get wet.

What to bring

- Your gold pan and a small shovel or garden trowel
- A bucket
- This workbook and a pencil
- Water shoes or boots, snacks, and plenty of water

At the creek: step by step

1. Walk the creek for 15 minutes before you touch a shovel. Go upstream and down. Just look.
2. Find the bends. On each one, look at the inside curve, the side with the gravel bar. That's where the water slows.
3. Find the head of each gravel bar, the upstream end where the bar begins. Heavy material drops there first.
4. Find the big boulders and look at the quiet water right behind them, on the downstream side.
5. Look for dark streaks in the gravel. That's black sand, and gold travels with it.
6. Look for bedrock showing at the edges of the creek and note any cracks. You'll come back for them on Trip 3.
7. Pick your 3 best spots, sketch them on the next page, and take one pan from each with your Tub Rehearsal motion.

Trip 2 Field Log

Your goal: A sketch of your creek with 3 spots marked, and a reason for each one.

My creek sketch

Draw the banks and an arrow for the direction the water flows. Mark the bends, gravel bars, big boulders and any bedrock. Put an X on your 3 spots.

flow direction: draw an arrow

My 3 spots

Spot	Why I picked it	What was in my pan?
A		
B		
C		

Josie's field note

In my first geology field course, the professor took our shovels away for a whole day and made us map a creek. I was so annoyed. By lunchtime I could see it: the gravel bars, the lazy water, the dark streaks. I've never looked at a river the same way since.

Ezra says: "Spend an hour looking before you spend a minute digging."

What I noticed: _____

☐ Trip 2 complete. I sketched my creek and sampled 3 spots.

Down to Bedrock

The Bedrock Rule: Gold sinks until it can't.

Where: exposed bedrock at the edge of your creek · **Time:** 2 to 3 hours

Why bedrock matters

Gold is so heavy that, over years of floods, it works its way down through the gravel until it hits solid rock. Then it has nowhere left to go. It settles into cracks, dips and potholes in the bedrock and stays there.

That's why old-timers always said to get down to bedrock. Where bedrock shows at the edge of a creek, you can reach those cracks with a screwdriver instead of a backhoe.

What to bring

- A crevice tool (a long flathead screwdriver or an old spoon works fine)
- A small stiff brush, your pan and a bucket
- Work gloves, and knee pads or a folded towel

At the creek: step by step

1. Find bedrock at the water's edge, ideally on an inside bend or where the creek has run over it.
2. Look for cracks that run across the current, not along it. Cross-cracks trap more.
3. Clear away the loose gravel sitting on top.
4. Scrape the crack out with your crevice tool, all the way to the bottom. The packed clay and fine material at the very bottom is what you want.
5. Brush everything from the crack into your pan.
6. Pan it with your Tub Rehearsal motion. Log every crack, including the empty ones.
7. Leave the spot tidy and put loose rocks back.

Trip 3 Field Log

Your goal: Clean and pan at least 5 bedrock cracks, and log every one.

Heads up: This is the trip where the Empty Pan Rule matters most. Plenty of cracks are empty. Log them and keep going.

My crack log

Crack	Across or along the current?	How deep?	Flakes
1			
2			
3			
4			
5			
6			
7			
8			

Josie's field note
My personal record is nine empty cracks in a row. I was ready to go home. The tenth one had eleven flakes in it, tucked into a little pocket of blue clay. They're still in a vial on my desk.

Ezra says: "Gold's lazy. It goes as low as it can and stays put."

What I noticed: _____

☐ Trip 3 complete. I cleaned and logged at least 5 cracks.

A Letter from Grandpa Ezra

Well now. If you're reading this, you've done three trips, and I'd wager at least one of them sent you home with an empty pan and a sore back.

Good. That means you're doing it right.

I've been at this since 1939. I've watched a lot of fellas show up at the creek with brand-new gear and big plans. Most of them were gone by the third weekend. Not because there wasn't gold. Because there wasn't any in the first few pans, and they took it personal.

The river don't care how you feel. It just keeps sorting. Every flood, it drops a little more gold in the same lazy places. The folks who find it are the folks who keep coming back to look.

So here's what I'll tell you, same as I told Josie when she was five and bawling over a coffee can lid. This exact moment, right here, halfway, when it feels like nothing's working? This is the whole game. Everybody who ever found anything stood right where you're standing now.

Turn the page. Do Trip 4.

Gold don't rush. Neither should you.

Ezra

Classify & Concentrate

The Bucket Rule: Move more dirt, not faster hands.

Where: your best spot from Trips 2 and 3 · **Time:** 2 to 3 hours

Why a bucket beats a pan

One pan is a sample. A bucket is a chance. The more material you move from a good spot, the better your odds. But panning big rocks is slow, and it's how beginners lose gold.

A classifier (a screen that sits on top of a bucket) fixes that. It sifts out the rocks so you only pan the fine, heavy material. And today you won't finish every pan. You'll stop at the heavy stuff, called concentrates, and save it for Trip 5.

What to bring

- A classifier screen (1/4 or 1/2 inch mesh) that fits a 5-gallon bucket
- Two 5-gallon buckets, a shovel and your pan
- A small jar with a lid for your concentrates

At the creek: step by step

1. Go to your best spot from Trip 2 or 3 and fill one bucket with material. Dig down toward bedrock if you can.
2. Set the classifier on the empty bucket and add a few scoops of material.
3. Pour creek water through it, or dunk and shake it in the creek, so the fines fall through into the bucket.
4. Glance at the rocks left on the screen. Any nuggets? It happens. Then toss them.
5. Pan the fines in batches. Stop each pan when you're down to a thin layer of dark, heavy material.
6. Tip those concentrates into your jar. Don't finish them here.
7. Time yourself. How many pans can you do in an hour?

Trip 4 Field Log

Your goal: Process one full 5-gallon bucket and save all your concentrates.

My bucket log

Bucket	Where it came from	Pans	Minutes	Cons saved?
1				
2				
3				
4				

Pans per hour today: _____

Josie's field note

The first time I used a classifier, I got through a whole bucket in the time it used to take me to do six pans. My knees practically wrote me a thank-you note.

Ezra says: "The river pays the fella who moves the most gravel, not the fella who wishes the hardest."

What I noticed: _____

☐ Trip 4 complete. I processed a full bucket and saved my concentrates.

Black Sand School

The Black Sand Rule: Where there's black sand, gold isn't far.

Where: at home, in a tub · **Time:** about 1 hour

Why black sand is your best clue

Black sand is mostly heavy iron minerals like magnetite. It isn't as heavy as gold, but it's heavier than almost everything else in the creek, so it travels with gold and settles in the same places.

That's why it ends up at the bottom of your pan, and why it's the best clue you have. Today you'll learn to separate gold from black sand gently, at home, where nothing can wash away.

What to bring

- Your concentrates jar from Trip 4
- A tub of water and your pan
- A strong magnet sealed inside a zip-top bag
- A snuffer bottle (a small squeeze bottle with a tube)
- A small glass vial filled with water

At the tub: step by step

1. Pan a small batch of concentrates in the tub, gently, until mostly black sand is left.
2. Leave a little water in the pan and tilt it slightly toward you.
3. Swirl slowly in a small circle and watch the back edge of the black sand. Gold lags behind and shows up as a little yellow tail.
4. Suck the flakes up with the snuffer bottle and squirt them into your vial.
5. Hold the bagged magnet just above the black sand to lift out the magnetite. Gold isn't magnetic, so it stays behind. Check what's left.
6. Save the leftover black sand in a jar. Fine gold hides in it, and you can re-pan it on any rainy day.

Trip 5 Field Log

Your goal: Recover every visible flake from your concentrates into your vial.

My black sand log

Batch	Black sand (little / some / lots)	Flakes found
1		
2		
3		
4		
5		
6		

My best black sand came from: _____

Josie's field note

The magnet trick blew my mind when I was nine. I spent that entire summer pulling black sand out of everything I could find, including the neighbor kid's sandbox. His mom was not a fan.

Ezra says: "Black sand is the river leaving you a note. Read it."

What I noticed: _____

☐ Trip 5 complete. Every visible flake is in my vial.

The Sample Line

The Sample Rule: One pan is a guess. Ten pans are a map.

Where: one inside bend or gravel bar on your creek · **Time:** 2 to 3 hours

Why we test in a pattern

Gold isn't sprinkled evenly. It collects in streaks and pockets, sometimes along a line the old-timers called the paystreak. The only way to find it is to test in a pattern and write everything down.

Prospectors count "colors," which just means the number of gold specks in one full pan. Professional geologists sample the same way, only with a drill instead of a pan.

What to bring

- Pan, shovel, bucket and classifier
- This workbook and a pencil
- 6 to 8 small flags or marked rocks to label your spots

At the creek: step by step

1. Choose one inside bend or gravel bar you sketched on Trip 2.
2. Mark 6 to 8 spots about 10 to 20 steps apart, from the head of the bar toward the tail, plus any bedrock cracks nearby.
3. At each spot, dig to the same depth and fill your pan the same amount. Fair tests only.
4. Pan each one all the way down and count the colors.
5. Write the count on your sketch and in the log right away. Don't trust your memory.
6. Look at the numbers. Is there a line where the counts go up? That's your streak.
7. Circle your best spot. That's where you'll go on Trip 7.

Trip 6 Field Log

Your goal: Log at least 6 sample pans and circle your best spot.

My sample line

Draw the bar and the flow arrow. Number your spots and write the color count next to each.

flow direction: draw an arrow

My sample log

Spot	Depth	Colors	Notes
1			
2			
3			
4			
5			
6			

My best spot: _____

Josie's field note

My professor's rule was simple: if it isn't written down, it didn't happen. I wrote "zero" so many times that first summer my notebook looked like a bowling scoresheet. But the page where the zeros stopped? That page was worth everything.

Ezra says: "I've dug a lot of holes in my life. The ones I wrote down are the ones that paid."

What I noticed: _____

☐ Trip 6 complete. I logged at least 6 pans and circled my best spot.

First Flake Day

The Stay Rule: Go back to your best spot, and stay there.

Where: your circled spot from Trip 6 · **Time:** half a day

Why today is different

Beginners hop around the creek chasing the next spot. Prospectors who find gold do the opposite. Once your sample line shows you where the numbers go up, you stay and work it.

Dig to bedrock, classify, concentrate, and finish carefully. Everything you've learned comes together today.

What to bring

- Everything: pan, shovel, classifier, two buckets, crevice tool
- Concentrates jar, snuffer bottle, vial, and the magnet in its bag
- A sewing pin for the tests, and your phone for a photo

At the creek: step by step

1. Go to your best spot from Trip 6.
2. Dig down toward bedrock and clean out any cracks you find.
3. Classify and concentrate at least one full bucket.
4. Finish your concentrates gently, at the creek or back home in the tub.
5. Snuff every flake into your vial.
6. Run the three tests on the next page.
7. Take a photo of your vial. You earned it.

Trip 7 Field Log

Your goal: At least one flake of gold in your vial, confirmed by all three tests.

The three tests

- ☐ **The Shade Test.** Tilt the vial or pan out of direct sun. Gold stays warm yellow from every angle. Pyrite and mica only flash when the light hits them.
- ☐ **The Pin Test.** Press one flake gently with a pin. Gold is soft and bends or flattens. Pyrite shatters. Mica splits into thin layers.
- ☐ **The Pan Test.** Gold stays put at the bottom of the pan. Mica is light and washes out easily.

My First Flake record

Date	
Creek	
My spot	
Weather	
Who was with me	
How many flakes	

What I'd tell myself back on Trip 1: _____

Josie's field note

My first real flake was the size of a poppy seed. I was five and Grandpa was seventy-seven, and we both yelled so loud that a fisherman downstream came running to see who'd gotten hurt.

Ezra says: "Josie's right. Everybody yells."

What I noticed: _____

- ☐ Trip 7 complete. I found my first flake.

Look How Far You Came

Copy your scores from *Before You Start*, then score yourself today.

What I rated	Before	After	Change
How confident I am about where to dig in a creek			
How good my panning is, without losing what I find			
How sure I am about where it's legal to pan			
How well I can tell real gold from fake			

That's evidence, not enthusiasm.

The Keep Three

Seven trips, seven rules. Pick the three that worked best for you and write them below. They're not lessons anymore. They're your habits.

1. _____
2. _____
3. _____

You read a river. It told you where the gold was, and you listened. That doesn't wear off. See you at the creek.

Josie

Read the river. It already knows where the gold is.

THE BACK OF THE BOOK

Your Field Kit

Pages to print, pin to the fridge, and carry to the creek. Plus four bonuses from me and Grandpa.

The Creek Card

Gear Checklist

Blank Sample Log

Bonus 1: Why Is My Pan Empty?

Bonus 2: Ezra's 10 Old-Timer Rules

Bonus 3: Prospector Talk

Bonus 4: The One-Day Plan

The Creek Card

The whole method on one page.

- | | |
|--------------------|---|
| 1. LOOK | Inside bends. Head of the gravel bar. Quiet water behind boulders. Bedrock cracks across the current. Black sand streaks. |
| 2. DIG | Go low. Bedrock if you can reach it. Scrape cracks all the way to the clay. |
| 3. CLASSIFY | Screen over a bucket. Check the big rocks, then toss them. Pan only the fines. |
| 4. PAN | Sink it. Break the clumps. Shake hard under water. Tilt, riffles away. Shake, tilt, wash. Repeat. |
| 5. FINISH | Save concentrates. Tilt toward you, swirl slow, find the yellow tail. Snuff it. Magnet in a bag for the black sand. |
| 6. TEST | Shade test. Pin test. Pan test. Gold glows, bends, and stays put. |

The seven rules

1. The Tub Rule: Lose it at home, so you never lose it in the river.
2. The Inside Bend Rule: Gold stops where the water gets lazy.
3. The Bedrock Rule: Gold sinks until it can't.
4. The Bucket Rule: Move more dirt, not faster hands.
5. The Black Sand Rule: Where there's black sand, gold isn't far.
6. The Sample Rule: One pan is a guess. Ten pans are a map.
7. The Stay Rule: Go back to your best spot, and stay there.

Gear Checklist

The essentials

- ☐ Gold pan with riffles
- ☐ Small shovel or garden trowel
- ☐ Two 5-gallon buckets
- ☐ Classifier screen that fits your bucket
- ☐ Crevice tool or long flathead screwdriver
- ☐ Snuffer bottle
- ☐ Small glass vials
- ☐ Jar with lid for concentrates
- ☐ This workbook and a pencil

If kids are coming

- ☐ Their own small pan
- ☐ Life jacket for any deeper water
- ☐ Extra dry clothes
- ☐ A jar just for their finds

Nice to have

- ☐ Small stiff brush
- ☐ Strong magnet in a zip-top bag
- ☐ Knee pads or a folded towel
- ☐ Bucket seat with a lid
- ☐ Small flags or marked rocks
- ☐ A pin for the Pin Test

Safety and comfort

- ☐ Waterproof gloves
- ☐ Dry clothes and socks in a bag
- ☐ Water and snacks
- ☐ Sun hat and sunscreen
- ☐ Charged phone, and someone who knows where you are
- ☐ Basic first aid kit
- ☐ Trash bag to pack out

COPY AS MANY AS YOU NEED

Blank Sample Log

Date: _____

Creek: _____

Spot	Depth	Colors	Notes
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			

BONUS 1

Why Is My Pan Empty?

Find your problem, then try the fix.

Problem	Likely cause	Try this
Nothing at all, pan after pan	You're digging where the water runs fast	Move to an inside bend, the head of a bar, or behind a boulder (Trip 2)
A few specks up top, then nothing	You're only sampling the surface	Dig deeper, toward bedrock (Trip 3)
Lots of black sand, no gold	Normal. Gold is much rarer than black sand	Keep sampling nearby. Black sand means you're in the right neighborhood (Trip 5)
I think I washed it out	Tilting too steep, or skipping the shake	Shake longer under water and re-level often (Trip 1)
Clay balls rolling around	The clay was never broken up	Break up every clump with your fingers under water before you pan
Shiny flakes that float away	Probably mica, not gold	Run the Pan Test and the Pin Test (Trip 7)
Gold in one spot, none a few steps away	Gold runs in streaks and pockets	Run a Sample Line and follow the numbers (Trip 6)
The creek is high and muddy	Spring runoff or recent rain	Come back when it drops. It's safer, and the gravel is easier to reach.

Ezra's 10 Old-Timer Rules

Collected over 87 years of cold water. Written down by Josie, because he won't.

1. Look an hour before you dig a minute.
2. Gold's lazy. Go low.
3. If it sparkles, be suspicious. If it glows, be happy.
4. Write down the zeros. They're half the map.
5. Fill your holes. The next fella deserves a creek, too.
6. Never trust a man who says he knows where the mother lode is. If he knew, he'd be digging it.
7. Cold water will beat you before the river does. Know when to go warm up.
8. Nobody ever got rich in one pan. Plenty did all right on flour gold they bothered to save.
9. Stay out of old tunnels. Every old-timer knows a name that didn't.
10. ***Gold don't rush. Neither should you.***

Ezra

Prospector Talk

Say these at the creek and people will think you've been at it for years.

Bedrock	The solid rock under the gravel. Gold sinks until it reaches it.
Black sand	Heavy dark minerals, mostly magnetite, that travel with gold.
Cheechako	An old Alaska and Yukon word for a newcomer. Everyone is one once.
Claim	A legal right to the minerals on a piece of ground, even on public land.
Classifier	A screen that sits on a bucket and sifts out the big rocks.
Colors	Specks of gold in a pan. "Five colors" means five specks.
Concentrates (cons)	The heavy material left near the bottom of the pan.
Creviceing	Cleaning gold out of cracks in bedrock.
Flour gold	Gold so fine it looks like dust.
Head of the bar	The upstream end of a gravel bar.
Mica	A shiny, flaky mineral that fools beginners. Light, and it floats away.
Paydirt	Gravel that holds enough gold to be worth working.
Paystreak	A line or layer where gold is concentrated.
Picker	A piece of gold big enough to pick up with your fingers.
Placer	Gold that's been freed from rock and moved by water.
Pyrite	Fool's gold. Brassy, brittle, and it flashes in the sun.
Riffles	The ridges on a pan or sluice that trap heavy material.
Skunked	Going home with nothing. It happens to everyone.
Snuffer bottle	A squeeze bottle with a tube for picking up flakes.
Sourdough	An experienced old-timer. The opposite of a cheechako.

The One-Day Plan

Only have one Saturday? Here's the whole book, squeezed into a day.

Time	What to do	From
The night before	Do a quick Tub Rehearsal with 10 BBs. Pack your gear with the checklist.	Trip 1
9:00	Arrive. Walk the creek for 15 minutes. Find bends, bar heads, boulders, bedrock.	Trip 2
9:30	Take one sample pan from each of your 3 best spots. Count colors.	Trips 2 and 6
10:30	Clean out 3 bedrock cracks near your best spot.	Trip 3
12:00	Lunch. Look at your notes. Pick one spot to stay at.	Trip 7
12:30	Classify and concentrate one full bucket from that spot.	Trip 4
2:30	Finish your concentrates slowly. Snuff every flake into a vial.	Trip 5
3:30	Run the three tests. Take a photo of your vial.	Trip 7
4:00	Fill your holes, pack out your trash, and head home happy.	Safety page

One day shows you it works. Seven trips make it yours.

My one-day notes: _____

Before You Close This Book

Josie and Ezra are AI-created characters. The people are invented. The geology, techniques and safety advice are real and were written to be accurate, but please double-check anything that matters with an official source.

This is not legal advice. Rules about where and how you can prospect vary by state, land type and season, and they change. Before you dig, check with the landowner, your local Bureau of Land Management field office, the Forest Service ranger district, or your state's natural resources or fish and wildlife agency.

This is not financial advice. Recreational panners usually find small amounts of gold. Do this for the creek, the skill and the yell, not for a paycheck.

Rivers can be dangerous. Cold water, fast currents, rising water and old mine workings can hurt or kill. You're responsible for your own safety and for anyone you bring along, especially kids. If in doubt, stay out.

If you're ever lost or hurt outdoors, call 911. Many remote areas have spotty signal, so tell someone your plan before you go.

Now go get your boots wet.

Josie